

# Ayaan M. Kazerouni

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Associate Professor  
Computer Science and Software Engineering  
Noyce School of Applied Computing  
California Polytechnic State University (Cal Poly)

## EDUCATION

|           |                      |                                            |
|-----------|----------------------|--------------------------------------------|
| 2015–2020 | PhD Computer Science | Virginia Tech, Blacksburg, VA              |
| 2011–2015 | BS Computer Science  | University of West Georgia, Carrollton, GA |

## EMPLOYMENT

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|-----------------|---------------------------------------------------------------------------------------------------------------|
| 09/2020–present | <b>Associate Professor</b><br><i>Computer Science and Software Engineering, Cal Poly, San Luis Obispo, CA</i> |
| 09/2020–08/2026 | <b>Assistant Professor</b><br><i>Computer Science and Software Engineering, Cal Poly, San Luis Obispo, CA</i> |
| 05/2020–08/2020 | <b>Summer Adjunct Faculty</b><br><i>Computer Science, Virginia Tech, Blacksburg, VA</i>                       |
| 06/2017–08/2017 | <b>Front-End Software Development Intern</b><br><i>Zappos.com, Las Vegas, NV</i>                              |
| 08/2015–04/2020 | <b>Graduate Research Assistant</b><br><i>Virginia Tech, Blacksburg, VA</i>                                    |

## PEER-REVIEWED PUBLICATIONS

Underlined author names indicate student authors (undergrad or MS).

### Journal Papers

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| TOCE 2025      | <i>The Benefits of Socially Responsible Computing in Early Computing Courses: A Multi-Institutional Study at Primarily-Undergraduate Hispanic-Serving Institutions.</i> ACM Transactions on Computing Education.<br><b>A. M. Kazerouni</b> , M. Lee, A. Hubbard Cheuoua, A. Gautam, S. Hooshmand, P. S. Inventado, E. E. Kang, J. Lehr, Y. Sun, K. A. Wortman, I. Yoon, Z. Wood. |
| J Chem Ed 2024 | <i>Recommendations for Improving End-User Programming Education: A Case Study with Undergraduate Chemistry Students.</i> ACS Journal of Chemical Education.<br><u>W. Fuchs</u> , A. McDonald, A. Gautam, <b>A. M. Kazerouni</b> .                                                                                                                                                |
| TOCE 2023      | <i>A Model of How Students Engineer Test Cases With Feedback.</i> ACM Transactions on Computing Education.<br><u>A. Shin</u> , <b>A. M. Kazerouni</b> .                                                                                                                                                                                                                          |
| JSS 2021       | <i>Fast and Accurate Incremental Feedback for Students' Software Tests Using Selective Mutation Analysis.</i> Journal of Systems and Software.<br><b>A. M. Kazerouni</b> , J. C. Davis, A. Basak, C. A. Shaffer, F. Servant, S. H. Edwards.                                                                                                                                      |

### Conference Papers

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ITiCSE 2026       | <i>Peer Networks and Instructor Resources as Complementary Support Systems in Six CS Classrooms (to appear).</i> ACM Conference on Innovation and Technology in Computer Science Education.<br><u>N. Ravetch</u> , <u>N. Nayman</u> , <u>J. Herberger</u> , T. Migler, <b>A. M. Kazerouni</b> . |
| Koli Calling 2025 | <i>What Topic Domains Interest Students in Socially Responsible Computing Coursework?.</i> Koli Calling International Conference on Computing Education Research.<br><b>A. M. Kazerouni</b> , Z. Agha, A. Hubbard Cheuoua, M. Lee, J. Lehr, I. Yoon, Z. Wood.                                   |
| SIGCSE 2025       | <i>Reflecting on Practices to Integrate Socially Responsible Computing in Introductory Computer Science Courses.</i> ACM Technical Symposium on Computer Science Education.<br>K. Wortman, A. Gautam, S. Hug, P. Salvador Inventado, <b>A. M. Kazerouni</b> , J. Lehr, K. Sood, Z. Wood.        |
| ASEE 2024         | <i>Socially Responsible Computing: Promoting Latinx Student Retention Via Community Engagement in Early Computer Science Courses.</i> ASEE Annual Conference and Exposition.                                                                                                                    |

D. M. Krum, Z. Wood, E. Kang, **A. M. Kazerouni**, J. L. Lehr, S. Hug, P. S. B. Inventado, F. Tang, I. Yoon, A. Kulkarni, Y. Sun, M. Beheshti, A. Gautam, A. Hubbard Cheuoua, S. Hooshmand, K. A. Wortman.

- SIGCSE 2024 *Community Action Computing: A Data-centric CS0 Course*. ACM Technical Symposium on Computer Science Education.  
**A. M. Kazerouni**, J. Lehr, Z. Wood.
- SIGCSE 2023 *Exploring the Impact of Cognitive Awareness Scaffolding for Debugging in an Introductory Computer Science Class*. ACM Technical Symposium on Computer Science Education.  
J. Lee, **A. M. Kazerouni**, C. Siu, T. Migler.
- Koli Calling 2021 *Patterns of Academic Help-Seeking in Undergraduate Computing Students*. Koli Calling Conference on Computing Education Research.  
A. Doebling, **A. M. Kazerouni**.
- SIGCSE 2021 *The Impact of Programming Project Milestones on Procrastination, Project Outcomes, and Course Outcomes: A Quasi-Experimental Study in a Third-Year Data Structures Course*. ACM Technical Symposium on Computer Science Education.  
C. A. Shaffer, **A. M. Kazerouni**.
- Koli Calling 2020 *Exploring the Bug Investigation Techniques of Intermediate Student Programmers*. Koli Calling Conference on Computing Education Research.  
R. S. Mansur, **A. M. Kazerouni**, S. H. Edwards, C. A. Shaffer.
- ITiCSE 2020 *ProgSnap2: A Flexible Format for Programming Process Data*. ACM Conference on Innovation and Technology in Computer Science Education.  
T. Price, D. Hovemeyer, K. Rivers, A. C. Bart, G. Gao, **A. M. Kazerouni**, B. Becker, A. Petersen, L. Gusukuma, S. H. Edwards, D. Babcock.
- ASE 2019 *Testing Regex Generalizability And Its Implications: A Large-Scale Many-Language Measurement Study*. IEEE/ACM International Conference on Automated Software Engineering.  
J. C. Davis, D. Moyer, **A. M. Kazerouni**, D. Lee.
- CompEd 2019 *The Relationship Between Practicing Short Programming Exercises and Exam Performance*. ACM Global Computing Education Conference.  
S. H. Edwards, Krishnan P. Murali, **A. M. Kazerouni**.
- SIGCSE 2019 *Assessing Incremental Testing Practices and Their Impact on Project Outcomes*. ACM Technical Symposium on Computer Science Education.  
**A. M. Kazerouni**, C. A. Shaffer, S. H. Edwards. **2nd Best Research Paper**.
- ICER 2017 *Quantifying Incremental Development Practices and Their Relationship to Procrastination*. ACM Conference on International Computing Education Research.  
**A. M. Kazerouni**, C. A. Shaffer, S. H. Edwards.
- ITiCSE 2017 *DevEventTracker: Tracking Development Events to Assess Incremental Development and Procrastination*. ACM Conference on Innovation and Technology in Computer Science Education.  
**A. M. Kazerouni**, C. A. Shaffer, T. S. Hall, S. H. Edwards.

#### Workshop Papers

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- EduVis 2026 *INKLING: Exploring How Students Design Visualizations with Integrated Sketching, Natural Language, and Code Editing*. 4th IEEE VIS Workshop on Visualization, Literacy, and Activities.  
S. Fisher, A. P. Wright, **A. M. Kazerouni**.

#### Abstracts and Posters

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- ICER 2019 *Toward Continuous Assessment of the Programming Process*. ACM Conference on International Computing Education Research-Doctoral Consortium.  
**A. M. Kazerouni**.
- SIGCSE 2019 *Student Debugging Practices and Their Relationships with Project Outcomes*. ACM Technical Symposium on Computer Science Education-Poster.  
**A. M. Kazerouni**, R. S. Mansur, S. H. Edwards, C. A. Shaffer.
- SIGCSE 2019 *Toward Continuous Assessment of the Programming Process (Abstract Only)*. ACM Technical Symposium on Computer Science Education-Student Research Competition.  
**A. M. Kazerouni**. **1st Place**.

## GRANT PROPOSALS

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### External Proposals

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- 2022 NSF #2216687: Collaborative Research: BPC-A: Socially Responsible Computing: Promoting Latinx Student Retention Via Community Engagement in Early CS Courses. Z. Wood, **A. M. Kazerouni**, J. Lehr, M. Beheshti, S. Hooshmand, P. S. Inventado, K. Sood, K. Wortman, E. E. Kang, D. Krum, Y. Sun, F. Tang, I. Yoon, A. Kulkarni, A. Gautam. Total amount: \$1.8M, Cal Poly share: \$513K.

### Internal Proposals

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Underlined author names indicate student authors (undergrad or MS).

- 2026 Noyce School of Applied Computing: Exploring the Design Space of Multimodal Visualization Authoring: Integrating Sketching, Code Editing, and Natural Language. **A. M. Kazerouni**, A. P. Wright. Amount: \$33,887.60.
- 2024 Summer Undergraduate Research Program: A Comparative Analysis of Software Test Adequacy Criteria. **A. M. Kazerouni**, J. B. Clements. 2 funded undergraduate students.
- 2022 Baker Koob Endowments: A Cost Effective Way of Measuring Software Test Success. A. Shin, **A. M. Kazerouni**. Amount: \$1400.
- 2022 Summer Undergraduate Research Program: Computer-supported Programmer Cognition. **A. M. Kazerouni**. 1 funded undergraduate student.

## TEACHING

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### Cal Poly

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|              |                                                              |
|--------------|--------------------------------------------------------------|
| 2022–2026    | CSC 123 Introduction to Community-Action Computing           |
| 2020–present | CSC 203 Project-based Object-oriented Programming and Design |
| 2023, 2024   | CSC 305 Individual Software Design and Development           |
| 2021         | CSC 307 Introduction to Software Engineering                 |
| 2021–present | CSC 313 Teaching Computing                                   |
| 2025–2026    | CSC 477 Scientific and Information Visualization             |
| 2021, 2023   | CSC 513 Computing Education Research and Practice (Graduate) |
| 2023         | CSC 590 Thesis Seminar (Graduate)                            |

### Virginia Tech

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|------------------|-----------------------------------------------------------|
| 2018, 2019, 2020 | CS 3114 Data Structures & Algorithms                      |
| 2015–2016        | CS 3114 Data Structures & Algorithms (Teaching Assistant) |

## GRADUATE ADVISING (MASTER'S THESES)

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### As committee chair

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|      |                             |                                                                                                                                              |          |
|------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2026 | <u>Parshana Sekhon</u>      | <i>Challenges and Opportunities of Mastery-Based Learning: A Student-Centered Analysis in an Upper-Division Programming Languages Course</i> | Cal Poly |
| 2026 | <u>Sinchana Shivaprasad</u> | <i>Pedagogical Perspectives of Practitioner-Turned-Instructors in Upper-Division Computer Science Courses</i>                                | Cal Poly |
| 2026 | <u>Nathan McCutchen</u>     | <i>Designing for Trust in Chat-Based Question Answering Systems: An Exchange-Based Retrieval Approach</i>                                    | Cal Poly |
| 2025 | <u>Ashton Alonge</u>        | <i>A Comparative Evaluation of Feedback Strategies for Enhancing Student Software Test Suite Writing Outcomes</i>                            | Cal Poly |
| 2024 | <u>Noah Ravetch</u>         | <i>Peer Collaborative Networks in Undergraduate Computing Classrooms</i>                                                                     | Cal Poly |
| 2024 | <u>Saurav Gupta</u>         | <i>A Comparative Study of the NPM, PyPI, Maven, and RubyGems Open-source Communities</i>                                                     | Cal Poly |

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| 2022 | <a href="#">Austin Shin</a>     | <i>Examining Introductory Computer Science Student Cognition When Testing Software Under Different Test Adequacy Criteria</i> | Cal Poly |
| 2022 | <a href="#">Will Fuchs</a>      | <i>Evaluating and Improving Domain-Specific Programming Education: A Case Study with Cal Poly Chemistry Courses</i>           | Cal Poly |
| 2022 | <a href="#">August Doebling</a> | <i>Patterns Of Academic Help-Seeking In Undergraduate Computing Students</i>                                                  | Cal Poly |
| 2021 | <a href="#">Kevin Yoo</a>       | <i>A Study of Non-computing Majors' Growth Mindset, Self-efficacy and Perceived CS Relevance in CS1</i>                       | Cal Poly |

#### As committee member

|      |                                |          |
|------|--------------------------------|----------|
| 2026 | <a href="#">Tymon Vu</a>       | Cal Poly |
| 2025 | <a href="#">Maggie Yang</a>    | Cal Poly |
| 2025 | <a href="#">Jamie Luna</a>     | Cal Poly |
| 2024 | <a href="#">Jordan Chew</a>    | Cal Poly |
| 2023 | <a href="#">Nikhil Sridhar</a> | Cal Poly |
| 2023 | <a href="#">Asmita Sharma</a>  | Cal Poly |
| 2023 | <a href="#">Martin Jiang</a>   | Cal Poly |
| 2022 | <a href="#">Jiwon Lee</a>      | Cal Poly |
| 2021 | <a href="#">Caleb Watts</a>    | Cal Poly |
| 2021 | <a href="#">Kylan Stewart</a>  | Cal Poly |
| 2021 | <a href="#">Justin Evans</a>   | Cal Poly |

## SERVICE

#### External Service

|                  |                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------|
| 2022–present     | Program Committee. <i>International Computing Education Research Conference (ICER)</i> .                   |
| 2024–present     | Program Committee. <i>Koli Calling Computing Education Research Conference (Koli Calling)</i> .            |
| 2022–2024        | Senior Program Committee. <i>SIGCSE Technical Symposium</i> .                                              |
| 2024             | Reviewer. <i>Computer Science Education</i> , Taylor & Francis.                                            |
| 2022–present     | Reviewer. <i>Transactions on Computing Education (TOCE)</i> .                                              |
| 2020, 2022, 2023 | Program Committee. <i>Conference on Innovation and Technology in Computer Science Education (ITiCSE)</i> . |
| 2020–2022        | Artefact Evaluation Program Committee. <i>Foundations of Software Engineering Conference (FSE)</i> .       |
| 2021, 2022       | Artefact Evaluation Program Committee. <i>Automated Software Engineering Conference (ASE)</i> .            |
| 2021             | Demonstrations Track Program Committee. <i>International Conference on Software Engineering (ICSE)</i> .   |
| 2018–2022        | Working Group Member. <i>CSSPLICE Programming Snapshot Data Working Group</i> .                            |
| 2019–2021        | Program Committee. <i>SIGCSE Technical Symposium</i> .                                                     |
| 2020             | Reviewer. <i>ASEE Computers in Education Journal</i> .                                                     |

#### Institutional Service

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|--------------|--------------------------------------------------------------------------------------------------------|
| 2021–present | CSSE tutoring centre advisor. <i>California Polytechnic State University</i> .                         |
| 2021–present | Club advisor: Hack4Impact. <i>California Polytechnic State University</i> .                            |
| 2024         | General Engineering Director Search Committee Member. <i>California Polytechnic State University</i> . |
| 2024         | CSSE Lecturer Search Committee Member. <i>California Polytechnic State University</i> .                |
| 2021         | CSSE Tenure-track Search Committee Member. <i>California Polytechnic State University</i> .            |
| 2017–2020    | President, Vice President, Treasurer/Cofounder. <i>CS Graduate Student Council, Virginia Tech</i> .    |

## HONORS AND AWARDS

- 2025 Cal Poly's nominee for the CSU Wang Family Award for Excellence in Teaching. *California Polytechnic State University (1 nominee per CSU campus)*
- 2024 Outstanding Reviewer Award. *Koli Calling Conference on Computing Education Research (6/80 reviewers)*
- 2024 Don and Paula Heye Annual Award for Outstanding Club Advisor (Hack4Impact). *College of Engineering, California Polytechnic State University*
- 2020 Graduate Student Service Award. *Department of Computer Science, Virginia Tech*
- 2019 2nd Best Paper Award, Research Track. *ACM SIGCSE Technical Symposium*
- 2018 1st Place in the SIGCSE Student Research Competition. *ACM SIGCSE Technical Symposium*
- 2015 2nd Place in the College of Math and Science Research Day. *University of West Georgia*
- 2014 Outstanding Honors Junior. *University of West Georgia*
- 2013 Outstanding Honors Sophomore. *University of West Georgia*